

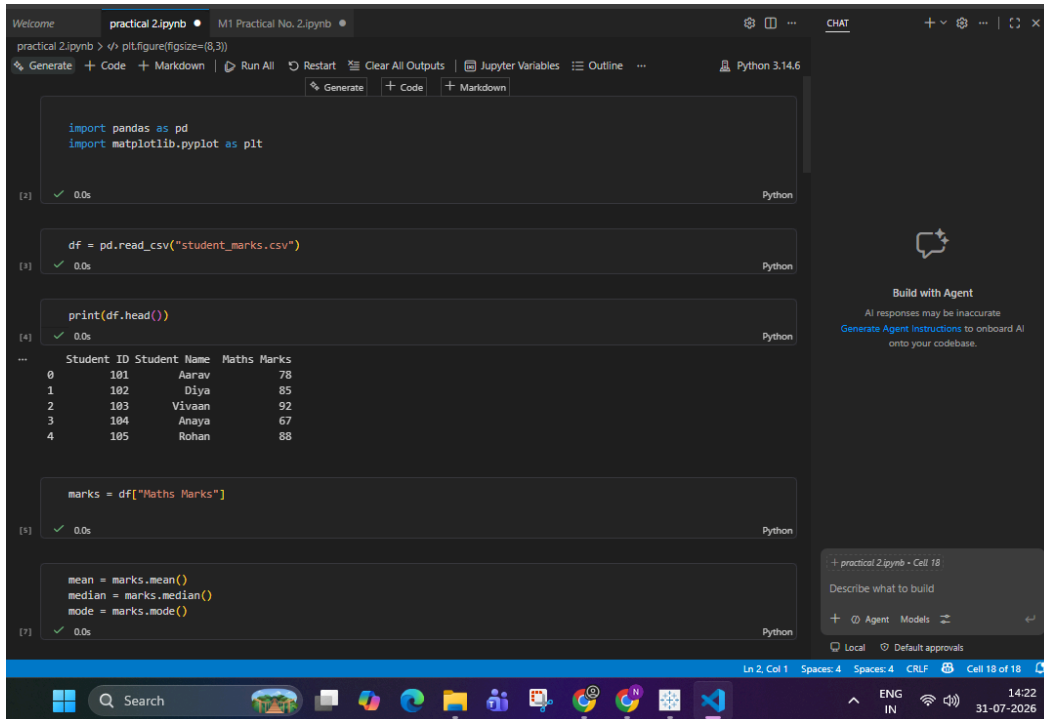
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T010

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
APPLIED BUSINESS ANALYTICS PRACTICAL
Practical – 2A&2B

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Class:TYIT	Batch:1
Date of Assignment:31-07-26	Date/Time of Submission:31-07-26

Q1.Obtain a Dataset and Calculate descriptive statistic(mean,median,mode,variance,etc)for a specific variable of interest

Code:-



```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student_marks.csv")

print(df.head())

marks = df["Maths Marks"]

mean = marks.mean()
median = marks.median()
mode = marks.mode()
```

The screenshot shows a Jupyter Notebook interface with the following content:

- Code cells with Python code for data analysis.
- Output of the first cell: A table showing the first 5 rows of the 'student_marks.csv' dataset.
- Output of the second cell: The first 5 rows of the 'Maths Marks' column.
- Output of the third cell: The mean, median, and mode of the 'Maths Marks' column.

	Student ID	Student Name	Maths Marks
0	101	Aarav	78
1	102	Diya	85
2	103	Vivaan	92
3	104	Anaya	67
4	105	Rohan	88

	Maths Marks
0	78
1	85
2	92
3	67
4	88

mean = 84.0
median = 85.5
mode = 85

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T010

The screenshot shows a JupyterLab environment with a file explorer on the left containing 'practical 2.ipynb' and 'student_marks.csv'. The main editor displays a Python script with the following code cells:

```
print("Mean:", mean)
print("Median:", median)
print("Mode:", mode.values)
```

Output: Mean: 82.4, Median: 83.5, Mode: [67 69 71 72 74 76 78 80 81 83 84 85 87 88 89 90 92 93 94 95]

```
print("Minimum :", marks.min())
```

Output: Minimum : 67

```
print("Maximum :", marks.max())
```

Output: Maximum : 95

```
print("Range :", marks.max() - marks.min())
```

Output: Range : 28

```
print("Variance :", marks.var())
```

Output: Variance : 76.56842105263156

The right sidebar features a 'Build with Agent' section with a warning that AI responses may be inaccurate and a button to 'Generate Agent instructions to onboard AI onto your codebase'.

The screenshot shows the continuation of the Python script in the JupyterLab environment. The code cells and their outputs are:

```
print("Standard Deviation :", marks.std())
```

Output: Standard Deviation : 8.758338339323319

```
print("Count :", marks.count())
```

Output: Count : 20

```
print("Sum :", marks.sum())
```

Output: Sum : 1648

```
print("\nDescriptive Statistics")
print(marks.describe())
```

Output: Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max
Maths Marks	20.000000	82.400000	8.758338	67.000000	75.500000	83.500000	89.250000	95.000000

Name: Maths Marks, dtype: float64

The right sidebar continues with the 'Build with Agent' section, showing a button to 'Describe what to build' and a 'practical 2.ipynb - Cell 14' entry.

Practical 2B

Q2. Create visualization(histogram box plot) to depicts the distribution of a variable and analyze its characteristics

